Assign Functional-Type Clusters

Description

Assigns taxa into functional types (FTs) using a pre-computed Gower dissimilarity matrix, a pre-fitted hierarchical clustering object, and a predetermined number of groups 'functional_type_group_count'. The number of groups is expected to have been chosen upstream via 'select_functional_type_group_count()'.

Usage

```
assign_functional_type_clusters(
  data_trait_table,
  trait_dissimilarity,
  hierarchical_clustering,
  functional_type_group_count,
  taxon_column = "taxon_name",
  verbose = TRUE
)
```

Arguments

`data_trait_table`	A data frame with one row per taxon. Must contain a character column (identified by 'taxon_column') holding taxon names. Must have at least 4 rows and 'nrow(data_trait_table)' must be strictly greater than 'functional_type_group_count'.
`trait_dissimilarity`	A 'dist' object produced by 'compute_trait_dissimilarity()'. Must inherit class '"dist"'. All distance values must be finite.
`hierarchical_clustering`	An 'hclust' object produced by 'fit_hierarchical_clustering()'. Must inherit class '"hclust"'.
`functional_type_group_count`	A single positive integer giving the number of functional-type groups to cut the dendrogram at. Must be at least 2 and less than 'nrow(data_trait_table)'.
`taxon_column`	A single character string naming the column in 'data_trait_table' that contains taxon names. Default: '"taxon_name"'.

`verbose`	A single logical. If 'TRUE' (default), prints the chosen 'functional_type_group_count' via 'cli::cli_inform()'.

Details

Algorithm:

1. Cut the dendrogram at 'functional_type_group_count' via 'stats::cutree()'.
2. Compute per-taxon silhouette widths via 'cluster::silhouette()'.
3. Return a tibble of taxon names, cluster assignments, and silhouette widths.

Distance computation (Inf -> NA, daisy, NaN -> 1.0) and hierarchical clustering are handled upstream by 'compute_trait_dissimilarity()' and 'fit_hierarchical_clustering()' respectively. FT-groups selection is handled upstream by 'select_functional_type_group_count()'. If 'nrow(data_trait_table) < 4' or 'functional_type_group_count >= nrow(data_trait_table)', the function aborts with an informative error.

Value

A tibble with columns:

taxon_name

: Taxon names (same values as 'data_trait_table[[taxon_column]]'). Column name is always '"taxon_name"' regardless of 'taxon_column'.

functional_type

: Integer label (1..'functional_type_group_count') giving the functional-type cluster assignment for each taxon.

silhouette_width

: Per-taxon silhouette width for the chosen 'functional_type_group_count' solution.

See Also

[compute_trait_dissimilarity()], [fit_hierarchical_clustering()], [select_functional_type_group_count()]